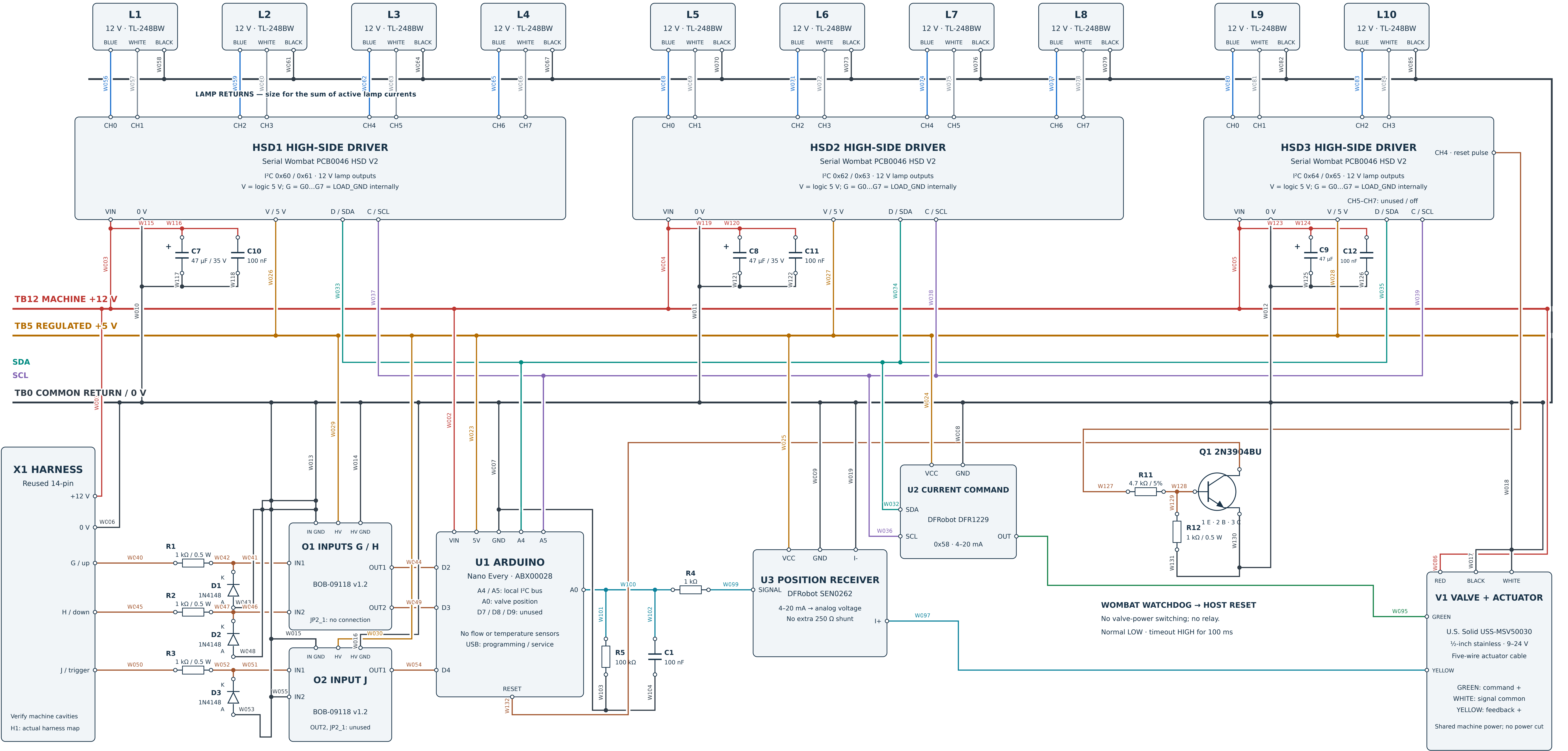


Read by terminal name, not by physical pin position.

Solid dot = joined wires. Gapped crossing = no connection. W-numbers match the connection register.

BENCH DESIGN — H1-H5 QUALIFICATION STILL REQUIRED

X1 labels are project signals, NOT the machine's numbered connector cavities.



CONTROL WITHOUT A FLOW METER

Valve power shares the machine supply; signal loops are direct.

Normal, Set Max, Flush and all ten lamps retain their controls.

An optional bucket / stopwatch calibration can map levels to flow.

A stored curve estimates flow only under similar water conditions.

No automatic flow sweep, live GPM or water-stop detection is provided.

Watchdogs recover the controller; power loss does not close the valve.

Live G/H/J with main power OFF: opto inputs draw about 9 mA each at 12 V.

Output-side HV = controller logic 5 V only. Bench-check the power-off case.

BUILD AND READING NOTES

- All 108 external wires are drawn. Shared trunks are joined terminals, not additional electronics.
- Component terminal positions are arranged for readability. Use the detailed sheets for physical pinouts.
- U1 GND1 = GND2 and RESET1 = RESET2; U2 GND = OUT_GND; HSD grounds are internally common.
- IC stays inside the enclosure at 100 kHz. Use only the U2 SDA/SCL pull-ups; open all HSD bus pull-up jumpers.
- HSD1 address jumpers open; HSD2 A1 closed; HSD3 A2 closed. All MAX_AMP jumpers open.
- HSD3 CH4 is the host-reset pulse, reserved for the watchdog. Never include it in lamp effects.
- Resistors: 1% except R11 5%. Ceramic capacitors: X7R, 50 V. C7–C9: 47 μ F, 35 V, 105 $^{\circ}$ C; observe polarity.
- Unused Nano pins: 3V3, A1/A2/A3/A6/A7, AREF, RX/TX, D5/D6/D7/D8/D9/D10/D11/D12/D13.
- No relay, coil regulator, flow meter or temperature probes. Codes 2 / 9 have no fitted sensors.
- Plumbing: hose $\rightarrow$ shutoff / strainer $\rightarrow$ valve $\rightarrow$ sprayer. Support the plumbing and keep electrical wiring dry.

Power loss does not spring-return the ball valve closed. Keep an accessible manual water shutoff.